

Next Generation SOLAR POWER TECHNOLOGY

The sun blesses the earth with enough energy in one hour— 4.3×10^{20} joules—to fulfill all of our energy needs for a year (4.1×10^{20} joules). Its giant bombardment of free and non-carbon-emitting energy has impelled scientists for decades to develop photon-electron converter solar cells that catch sunlight and convert it directly into electricity.

Silicon solar cells are in use since the beginning of the space programme and now dominate the industry, with worldwide shipments of solar products expected to have totaled \$41.9 billion in 2016. Silicon solar cell technology has captured 90 per cent of the solar market, but scientists around the world are working to find a way to make them more efficient, affordable, stable and reliable.

The typical silicon wafer solar panel is power rated using sunlight at zenith (climax or high point), which provides a radiance of just over one kilowatt per square metre at sea level. This radiant energy comprises 445 watts of visible light component, 527 watts of IR radiation and 32 watts of UV radiation. Wafer-type silicon solar cells available today can process only visible sunlight. Thus 56 per cent of the available energy to be converted into electricity is unutilised with silicon solar cells.

The pursuit of lower-cost and more efficient solar cells has led to the search of new materials. In the past several years, several emerging photovoltaic technologies were the focus of R&D effort in the academia and industry. These third-generation solar cells do not require custom-made, high-priced, complicated processing equipment, as also expensive silicon or rare earth elements such as indium.

According to National Renewable Energy Laboratory (NREL), which is an international organisation for certification of solar cell efficiency, multicrystalline silicon cells are 21 per cent efficient and expensive, while single-crystal silicon cells have efficiency of around 25 per cent. The highest efficiency of 46 per cent has been recorded for gallium arsenide-based multijunction solar cells but at excessively high cost. The bright CIGS photovoltaic cells are 22.6 per cent efficient. One of the potential application of these solar cells was envisaged for window glasses. Thin films of semitransparent photovoltaic modules used in place of window glasses on a building will allow people inside to see through the window while appearing as tinted glass from the outside.

The promising emerging photovoltaic technologies, called third generation PV, are organic photovoltaics, dye-sensitised solar cells, perovskite photovoltaics and inorganic quantum-dot solar cells. Unlike traditional silicon solar cells, these represent transformative technologies with great potential for extremely high-throughput manufacturing at very low cost, low environmental impact, mechanical flexibility and molecular tailorability while using non-toxic, earth-abundant materials with short energy payback times. This makes them suitable for applications beyond rooftop and solar-farm panels. The fundamental advantage of these thin-film technologies comes in the form of the amount of material required for fabrication.

However, the lower price advantage of these emerging photovoltaics is offset by their lower efficiency. The power conversion efficiency (a ratio of light energy in to electrical energy out) of dye-sensitised solar cells, organic photovoltaics and quantum-dot cells today sits at roughly 12 per cent. Hence these emerging photovoltaic technologies are applicable only in niche markets that require light weight, flexibility and variable-angle performance, such as in consumer electronics. Some are



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Third-generation solar cells, whose production has started, hold a bright future

COMPARISON OF THIRD-GENERATION PV TECHNOLOGIES

3rd generation PV technology	Maximum efficiency certified by NREL	Commercial status
Organic photovoltaics	11.5 per cent	Advanced demonstration, some products for sale
Dye-sensitised solar cells	11.9 per cent	Advanced demonstration, some products for sale
Quantum-dot photovoltaics	13.4 per cent	Under development, no products
Perovskite solar cells	22.1 per cent (not stabilised)	Under development, no products